

Storing large volumes of data through traditional systems is very expensive. So for most of the critical data, organizations, relying on traditional systems, have not had the capability to save it, organize it, and utilize it or analyze its benefits because of the storage costs. The ever increasing volume of data, the unstoppable velocity of the data that is being generated, and the complexity of working with unstructured data as well as the costs have kept these organizations from leveraging the details of the data and that pulled them back from making good business decisions in an

ever-changing competitive environment. During that time, Google was evolving and it had the challenge to be able to rank the Internet. For this, it had to design a new way of solving the problem. It designed a platform for itself that offered inexpensive storage, was easily scalable, could access data very fast and store structured, semi-structured, and unstructured data to make it easy to analyze the data and find a relationship among the data.

MONITOR FOOTBALL PLAYER ACTIONS



The National Football League (NFL) installed RFID data sensors in the shoulder pads of each player to collect location data which helped to keep track and analyze player acceleration and speed, which was tested during 2015 Pro Bowl event.

Introduction to Hadoop

Apache™ Hadoop® is an open source software based on Big Data technology, written in Java, that enables processing of a very large amount of data by distributing these sets of data across different cluster (thousands of nodes) of servers. It is designed to scale up from a single machine to thousands of machines, and each of these machines offer local computation and storage. Hadoop was created by Doug Cutting and Mike Cafarella in 2005.

The Apache Hadoop framework is composed of the following four modules:

- ♦ **Hadoop Common:** It is also known as the Hadoop Core. It refers to the collection of common libraries and utilities that support other Hadoop modules. It is an essential module of the Hadoop Framework and is designed in such a manner that hardware failures are automatically